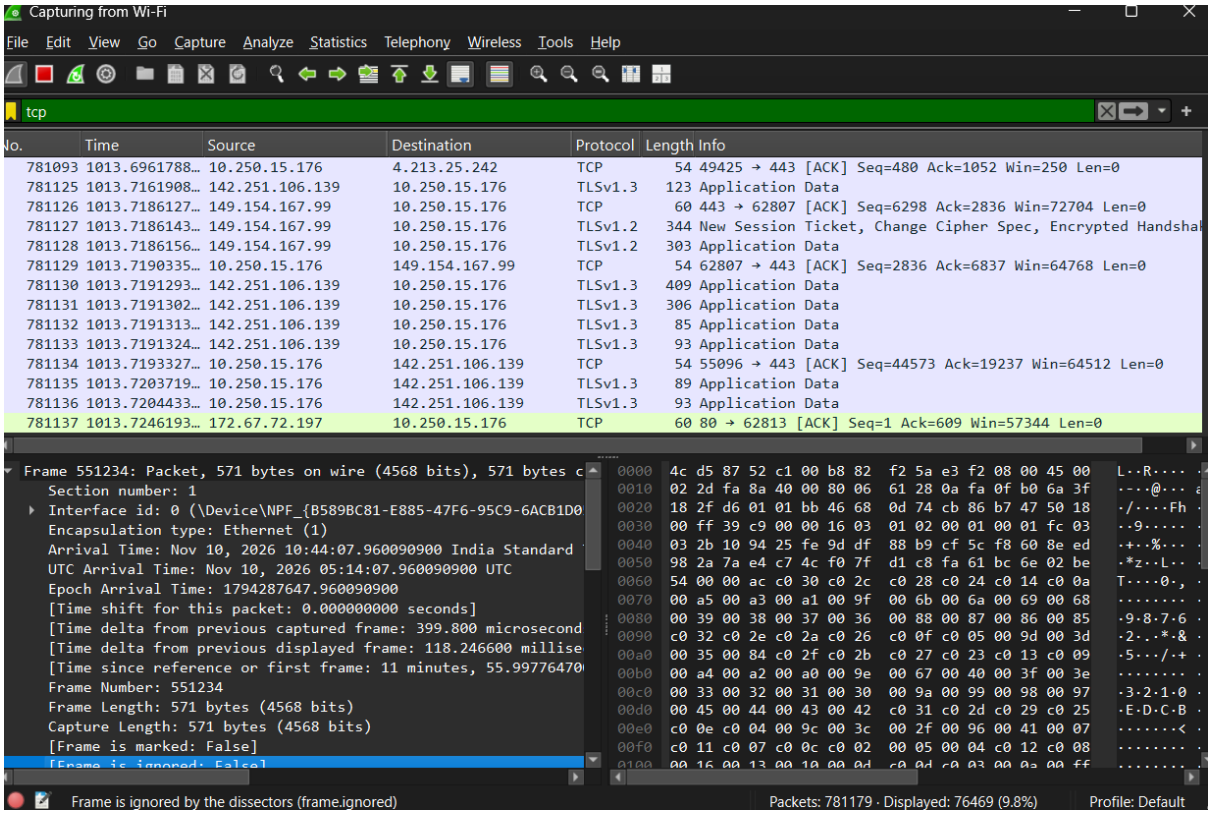
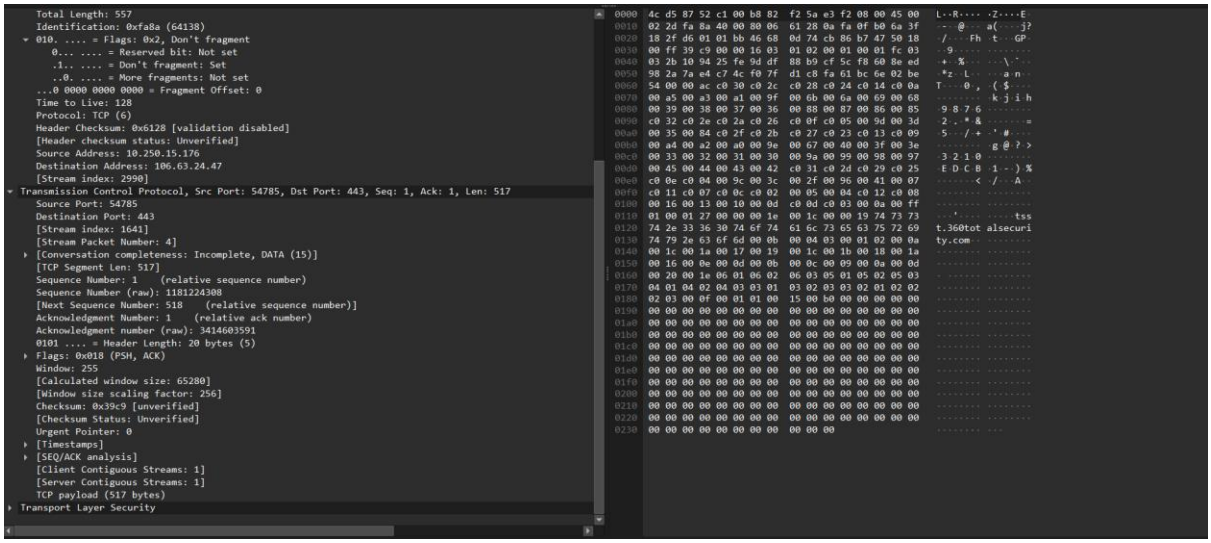
Skill - 1

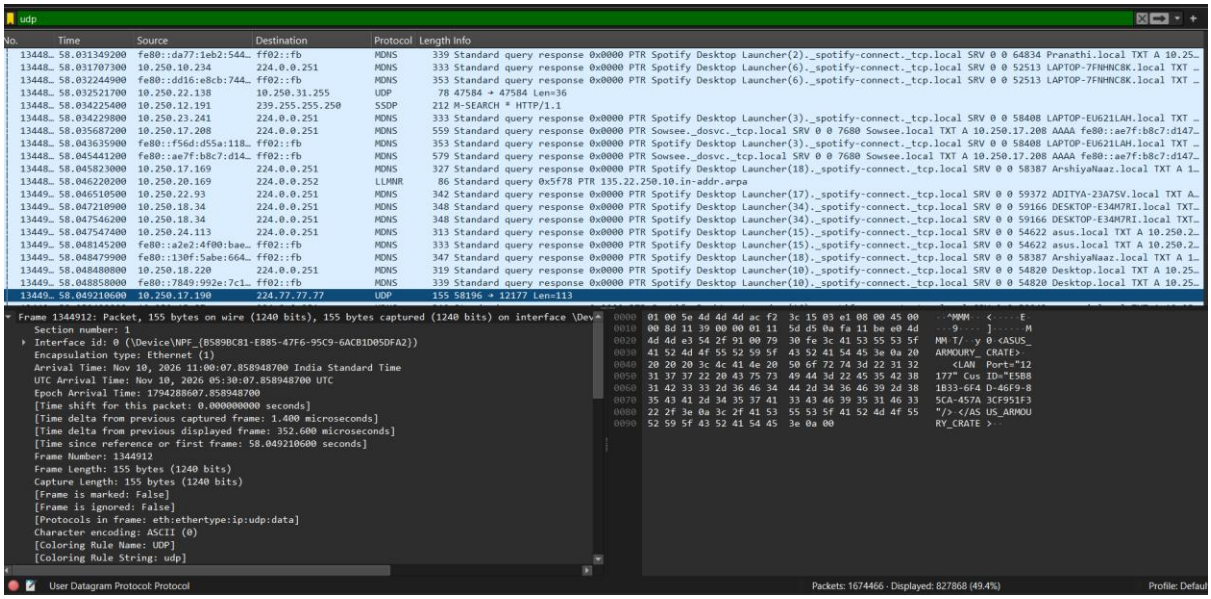
Encapsulation Builder: Analyze the overhead introduced by protocol headers and determine how encapsulation affects network efficiency.

Aim: To capture network packets and understand how Ethernet, IP, and TCP/UDP headers are added during data transmission.

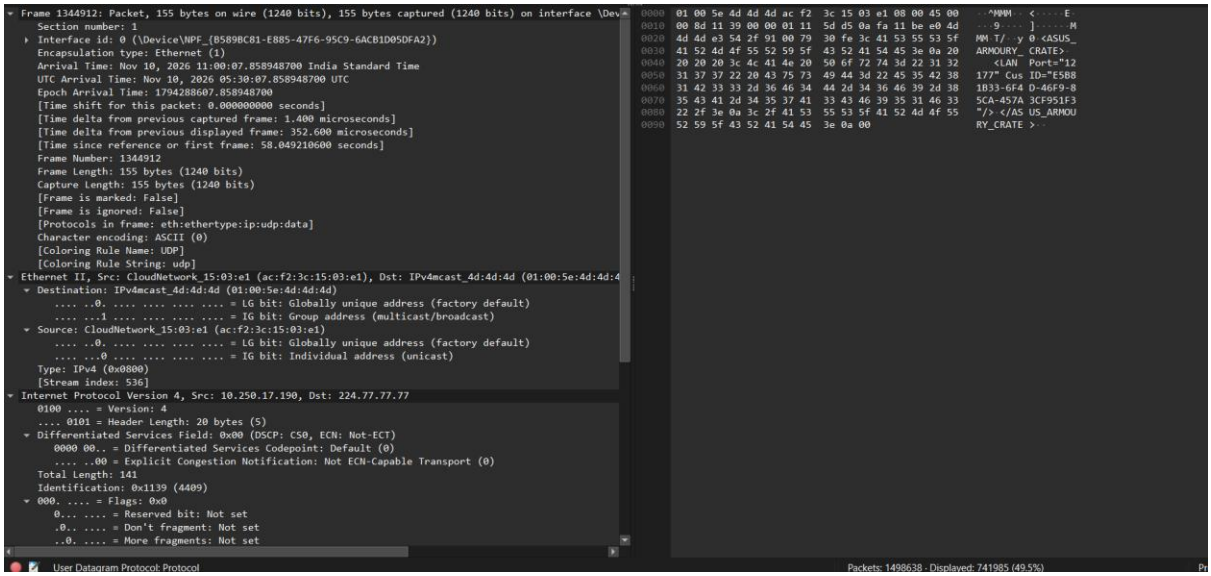


```
Frame 551234: Packet, 571 bytes on wire (4568 bits), 571 bytes captured (4568 bits) on interface \D
Ethernet II, Src: WNC_5a:e3:f2 (b8:82:f2:5a:e3:f2), Dst: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
  Destination: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
      .... ..0. .... .... .... .... = LG bit: Globally unique address (factory default)
      .... ...0 .... .... .... .... = IG bit: Individual address (unicast)
  Source: WNC_5a:e3:f2 (b8:82:f2:5a:e3:f2)
  Type: IPv4 (0x0800)
  [Stream index: 116]
Internet Protocol Version 4, Src: 10.250.15.176, Dst: 106.63.24.47
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
     0000 00.. = Differentiated Services Codepoint: Default (0)
     .... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
  Total Length: 557
  Identification: 0xfa8a (64138)
  010. .... = Flags: 0x2, Don't fragment
     0... .... = Reserved bit: Not set
     .1.. .... = Don't fragment: Set
     ..0. .... = More fragments: Not set
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 128
  Protocol: TCP (6)
  Header Checksum: 0x6128 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 10.250.15.176
  Destination Address: 106.63.24.47
  [Stream index: 2990]
Transmission Control Protocol, Src Port: 54785, Dst Port: 443, Seq: 1, Ack: 1, Len: 517
  Source Port: 54785
  Destination Port: 443
  [Stream index: 1641]
  [Stream Packet Number: 4]
  [Conversation completeness: Incomplete, DATA (15)]
  [TCP Segment Len: 517]
  Sequence Number: 1    (relative sequence number)
  Sequence Number (raw): 1181224308
  [Next Sequence Number: 518    (relative sequence number)]
  Acknowledgment Number: 1    (relative ack number)
  Acknowledgment number (raw): 3414603591
  0101 .... = Header Length: 20 bytes (5)
```





Frame 1344912: Packet, 155 bytes on wire (1240 bits), 155 bytes captured (1240 bits) on interface \Dev
Section number: 1
Interface id: 0 (\Device\NPF_{B589BC81-E885-47F6-95C9-6ACB1D05DFA2})
Encapsulation type: Ethernet (1)
Arrival Time: Nov 10, 2026 11:00:07.858948700 India Standard Time
UTC Arrival Time: Nov 10, 2026 05:30:07.858948700 UTC
Epoch Arrival Time: 1794288607.858948700
[Time shift for this packet: 0.000000000 seconds]
[Time delta from previous captured frame: 1.400 microseconds]
[Time delta from previous displayed frame: 352.600 microseconds]
[Time since reference or first frame: 58.049210600 seconds]
Frame Number: 1344912
Frame Length: 155 bytes (1240 bits)
Capture Length: 155 bytes (1240 bits)
[Frame is marked: False]
[Frame is ignored: False]
[Protocols in frame: eth:ethertype:ip:udp:data]
Character encoding: ASCII (0)
[Coloring Rule Name: UDP]
[Coloring Rule String: udp]
Ethernet II, Src: CloudNetwork_15:03:e1 (ac:f2:3c:15:03:e1), Dst: IPv4mcast_4d:4d:4d (01:00:5e:4d:4d:4
Destination: IPv4mcast_4d:4d:4d (01:00:5e:4d:4d:4d)
.... ..0. .... = LG bit: Globally unique address (factory default)
....1 .... = IG bit: Group address (multicast/broadcast)
Source: CloudNetwork_15:03:e1 (ac:f2:3c:15:03:e1)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Type: IPv4 (0x0800)
[Stream index: 536]
Internet Protocol Version 4, Src: 10.250.17.190, Dst: 224.77.77.77
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
0000 00.. = Differentiated Services Codepoint: Default (0)
.... ..00 = Explicit Congestion Notification: Not ECN-Capable Transport (0)
Total Length: 141
Identification: 0x1139 (4409)
000. .... = Flags: 0x0
0... .... = Reserved bit: Not set
.0.. .... = Don't fragment: Not set
..0. .... = More fragments: Not set



Parameter	TCP Packet	UDP Packet
Frame Length	571 bytes	155
Ethernet Header Length	14	14
IP Header Length	20	20
Transport Header Length	20	8
IP Total Length	557	141
Transport Header Length	20	8
Payload Length	517	113